



CTI and Associates, Inc.

12482 Emerson Drive Brighton, MI 48116 248.486.5100 248.486.5050 Fax

March 22, 2007

Mr. Mike Bray
Unit Chief
Permits Section
Michigan Department of Environmental Quality - Water Bureau
P.O. Box 30457
Lansing, Michigan 48909

RECEIVED

MAR 27 2007

W.D.-SURFACE WATER
PERMITS SECTION

Subject: Effluent Limits for Planning Purposes
Wexford County Landfill, Cedar Creek Township,
Wexford County, Michigan

Dear Mr. Bray:

On behalf of the Wexford County Department of Public Works, (DPW), CTI and Associates, Inc. (CTI) requests National Pollutant Discharge Elimination System (NPDES) permit related effluent limits for exploring and evaluating alternatives for leachate treatment at the Wexford County Landfill (WCL). The WCL is an existing Type II sanitary landfill located in Cedar Creek Township, south of the City of Manton, Wexford County, Michigan. The landfill is operated by the Wexford County DPW and is located at 990 North US 131, Manton, Michigan 49663.

The site is bordered by private land to the north and west, and by undeveloped public land to the south and east of the site. WCL consists of approximately 187.4 acres within the property boundary. A facility map is shown in Figure 1.

Currently, leachate from the facility is transported by tanker truck(s) and treated at the City of Cadillac WWTP (Cadillac WWTP). WCL is the only "bulk" disposal customer of the Cadillac WWTP and the relatively high concentration of ammonia in the leachate poses a significant operational hardship for their system. To mitigate this problem, the Cadillac WWTP has placed a limit on the amount of leachate they will accept from the WCL. WCL has a total working storage capacity of approximately 145,000 gallons. If Cadillac WWTP continues to impose the daily discharge limit the landfill they will need to haul the leachate out of the County to the Grand Rapids WWTP.

As the Cadillac WWTP is not designed to optimally handle landfill leachate, providing on-site state of the art treatment (Two Stage SBR) at the Wexford County Landfill will benefit all parties



involved. In addition, the conventional biological pollutant concentrations in leachate and residential septage are similar. To our knowledge, there is no licensed septage receiving facility located in Wexford County. The treatment technology to adequately treat both of these wastes would require minor adjustments.

The following information will assist you in developing the effluent limits for this planning purpose.

Average Flow: 100,000 gallons per day
Peak Flows: 300,000 gallons per day
Daily Max Flows: 200,000 gallons per day

Outfall Location: General Permit # MIS111450 Outfall
NE1/4, NE1/4, Section 34, Town 23 N, W, Wexford County to the
Manton Creek via Unnamed creek.

Enclosed for you review, please find results from an influent test recently performed by an independent laboratory.

We look forward to receiving the effluent limits to develop an environmentally responsible plan for the Wexford County Landfill.

If there are questions or concerns, please do not hesitate to contact Mr. John Divozzo at 213.775.0155 or myself at 248.467.7308.

Very truly yours,

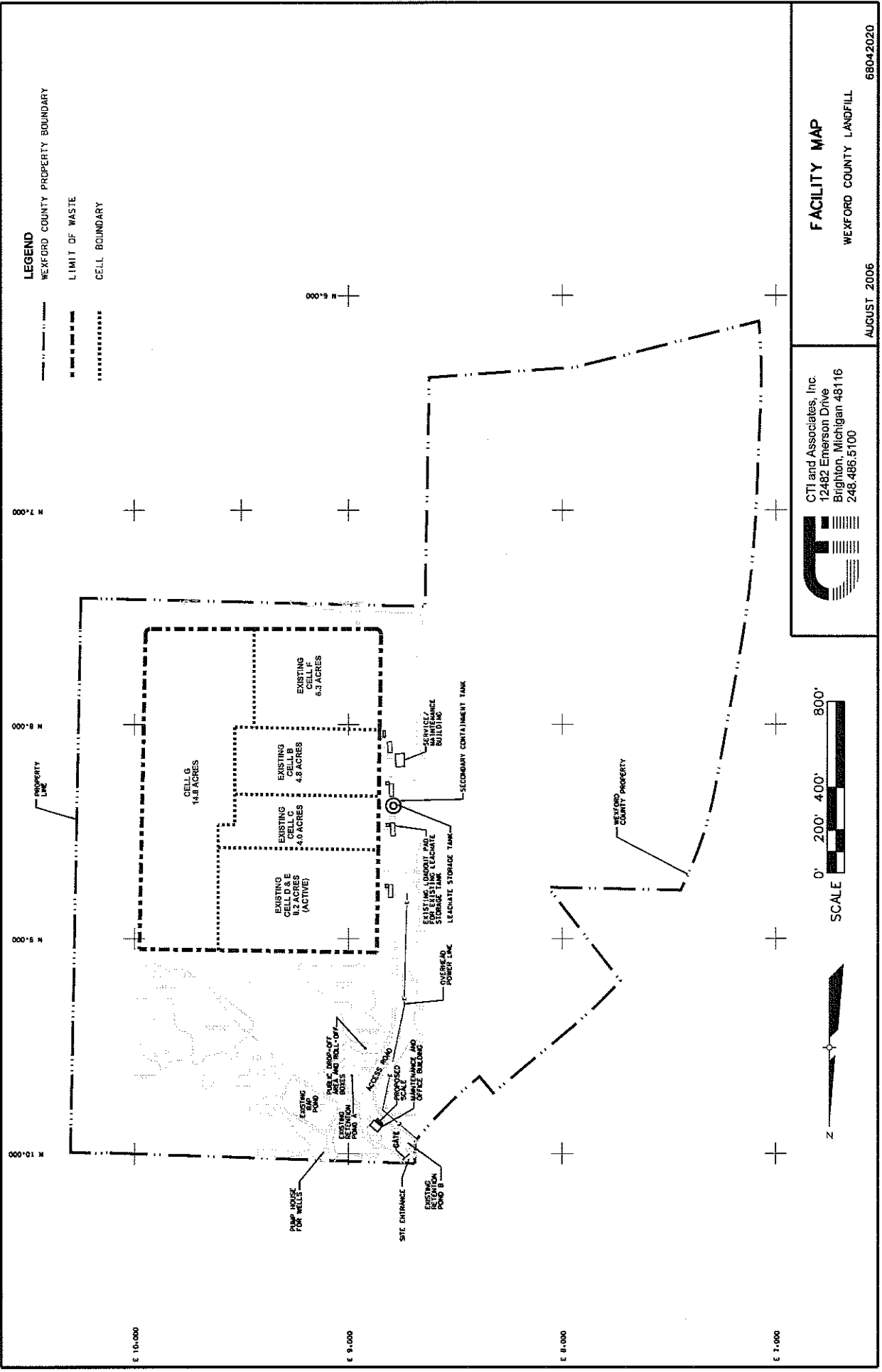
CTI and Associates, Inc.

A handwritten signature in black ink that reads 'Beth A. Benoit'. The signature is written in a cursive, flowing style.

Beth A. Benoit
Client Manager

Attachments

Cc: John Divozzo, Director Public Works
Rich Leszcz, Landfill Manager
Project File



CTI and Associates, Inc.
12482 Emerson Drive
Brighton, Michigan 48116
248.486.5100

FACILITY MAP

WEXFORD COUNTY LANDFILL

AUGUST 2006

68042020

Wexford County Landfill Leachate Sampling

Total Metals (ug/L)	Sample Identification				
	Leachate Tank 1st Quarter 2006	Leachate Tank 2nd Quarter 2006	Leachate Tank 3rd Quarter 2006	Leachate Tank 4th Quarter 2006	Leachate Tank January 2007
Antimony	< 2.0	2.9	< 2.0	< 2.0	NA
Arsenic	6.7	5.3	5.0	5.2	NA
Barium	410	620	880	540	NA
Beryllium	< 1.0	< 1.0	< 1.0	< 1.0	NA
Boron	4,600	4,800	5,400	4,600	NA
Cadmium	< 0.20	< 0.20	< 0.20	< 0.20	NA
Calcium	150,000	270,000	240,000	160,000	NA
Chromium	7.0	9.0	6.0	5.0	NA
Cobalt	11	< 10	< 10	< 10	NA
Copper	5.9	< 2.0	18	5.6	NA
Iron	3,000	30,000	58,000	19,000	NA
Lead	6.9	< 1.0	< 1.0	< 5.0	NA
Magnesium	140,000	170,000	180,000	140,000	NA
Manganese	2,300	3,200	1,300	550	NA
Molybdenum	NA	NA	NA	NA	77
Nickel	26	37	34	30	NA
Potassium	990,000	1,000,000	780,000	730,000	NA
Selenium	< 5.0	< 5.0	< 5.0	< 5.0	NA
Silver	< 0.50	< 0.50	< 0.50	< 0.50	NA
Sodium	720,000	710,000	730,000	540,000	NA
Thallium	< 1.0	< 1.0	< 1.0	< 1.0	NA
Vanadium	< 10	< 10	< 10	< 10	NA
Zinc	< 10	< 10	43	20	NA

Notes:

ug/L = micrograms per liter.

ND = Not detected above MDL

NS = Not sampled for analyte

Compounds which exceeded Method Detection Limits (MDL)

are indicated in bold

The laboratory report lists all parameters analyzed.

Wexford County Landfill

Leachate Sampling

VOCs (ug/L)	Sample Identification			
	Leachate Tank 1st Quarter 2006	Leachate Tank 2nd Quarter 2006	Leachate Tank 3rd Quarter 2006	Leachate Tank 4th Quarter 2006
Acetone	160	< 1,000	< 1,000	200
Benzene	5.4	< 10	6.8	5.3
Bromobenzene	<1.0	< 10	<1.0	<1.0
Bromochloromethane	<1.0	< 10	<1.0	<1.0
Bromodichloromethane	<1.0	< 10	<1.0	<1.0
Bromoform	<1.0	< 10	<1.0	<1.0
Bromomethane	<1.0	< 10	<1.0	<1.0
2-Butanone	200	1,300	890	300
n-butylbenzene	<1.0	< 10	<1.0	<1.0
sec-butylbenzene	<1.0	< 10	<1.0	<1.0
tert-butylbenzene	<1.0	< 10	<1.0	<1.0
Carbon disulfide	<50	< 500	<50	<50
Carbon tetrachloride	<1.0	< 10	<1.0	<1.0
Chlorobenzene	<1.0	< 10	<1.0	<1.0
Chloroethane	<1.0	< 10	<1.0	<1.0
Chloroform	<1.0	< 10	<1.0	<1.0
Chloromethane	<1.0	< 10	<1.0	<1.0
2-Chlorotoluene	<1.0	< 10	<1.0	<1.0
4-Chlorotoluene	<1.0	< 10	<1.0	<1.0
Dibromochloromethane	<1.0	< 10	<1.0	<1.0
1,2-Dibromo-3-chloropropane	<1.0	< 10	<1.0	<1.0
1,2-Dibromoethane	<1.0	19	<1.0	<1.0
Dibromomethane	<5.0	< 50	<5.0	<5.0
1,2-Dichlorobenzene	<1.0	< 10	<1.0	<1.0
1,3-Dichlorobenzene	<1.0	< 10	<1.0	<1.0
1,4-Dichlorobenzene	<1.0	< 10	<1.0	<1.0
trans-1,4-Dichloro-2-butene	<1.0	< 10	<1.0	<1.0
Dichlorodifluoromethane	<1.0	< 10	<1.0	<1.0
1,1-Dichloroethane	<1.0	< 10	<1.0	<1.0
1,2-Dichloroethane	7.6	< 10	7.1	7.1
1,1-Dichloroethene	<1.0	< 10	<1.0	<1.0
cis-1,2-Dichloroethene	5.6	< 10	10	10
trans-1,2-Dichloroethene	<1.0	< 10	<1.0	<1.0
cis-1,3-Dichloropropene	<1.0	< 10	<1.0	<1.0
trans-1,3-Dichloropropene	<1.0	< 10	<1.0	<1.0
1,2-Dichloropropane	<1.0	< 10	<1.0	<1.0
1,3-Dichloropropane	<1.0	< 10	<1.0	<1.0
2,2-Dichloropropane	<1.0	< 10	<1.0	<1.0
1,1-Dichloropropene	<1.0	< 10	<1.0	<1.0
Diethyl ether	11	< 100	12	11
Ethylbenzene	15	19	16	12
Hexachlorobutadiene	<5.0	< 50	<5.0	<5.0
Hexachloroethane	<5.0	< 50	<5.0	<5.0

Wexford County Landfill

Leachate Sampling

VOCs (ug/L)	Sample Identification			
	Leachate Tank 1st Quarter 2006	Leachate Tank 2nd Quarter 2006	Leachate Tank 3rd Quarter 2006	Leachate Tank 4th Quarter 2006
2-Hexanone	<50	< 500	<50	<50
Idomethane	<1.0	< 10	<1.0	<1.0
Isopropylbenzene	5.2	< 10	5.0	4.4
4-Isopropyltoluene	<1.0	< 10	<1.0	<1.0
4-Methyl-2-pentanone	<50	< 500	64	<50
Metyl tert-butyl ether (MTBE)	6.8	< 40	7.3	4.6
Methylene chloride	<5.0	< 50	<5.0	<5.0
n-propylbenzene	<1.0	< 10	<1.0	<1.0
Styrene	<1.0	< 10	<1.0	<1.0
1,1,1,2-Tetrachloroethane	<1.0	< 10	<1.0	<1.0
1,1,2,2-Tetrachloroethane	<1.0	< 10	<1.0	<1.0
Tetrachloroethene	1.6	< 10	<1.0	<1.0
Toluene	2.2	21	27	17
1,2,3-Trichlorobenzene	<5.0	< 50	<5.0	<5.0
1,2,4-Trichlorobenzene	<5.0	< 50	<5.0	<5.0
1,1,1-Trichloroethane	<1.0	< 10	<1.0	<1.0
1,1,2-Trichloroethane	<1.0	< 10	<1.0	<1.0
Trichloroethene	<1.0	< 10	<1.0	<1.0
Trichlorofluoromethane	<1.0	< 10	<1.0	<1.0
1,2,3-Trichloropropane	<1.0	< 10	<1.0	<1.0
1,2,4-Trimethylbenzene	2.8	< 50	2.0	1.7
1,3,5-Trimethylbenzene	<1.0	< 10	<1.0	<1.0
Vinyl chloride	<1.0	< 10	<1.0	<1.0
Xylenes	10.9	< 30	15.5	11.6

Notes ug/L = micrograms per liter.
 ND = Not detected above MDL.
 NS = Not sampled for analyte
 Compounds which exceeded Method Detection Limits (MDL)
are indicated in bold.
 The laboratory report lists all parameters analyzed.
 VOCs = Volatile organic compounds
 N/A=Paramter was not analyzed during this event

Wexford County Landfill Leachate Sampling

Inorganic Indicators (mg/L)	Sample Identification			
	Leachate Tank 1st Quarter 2006	Leachate Tank 2nd Quarter 2006	Leachate Tank 3rd Quarter 2006	Leachate Tank 4th Quarter 2006
Ammonia	110	170	190	200
Bicarbonate	2,700	2,700	3,000	2,200
Carbonate	< 2.0	< 2.0	< 2.0	< 2.0
Chemical Oxygen Demand	620	1,700	1,700	580
Chloride	1,100	980	1,100	720
Cyanide	< 5.0	< 5.0	< 5.0	< 5.0
Nitrate	< 0.10	< 0.10	< 0.10	< 0.10
Nitrite	< 0.05	< 0.05	< 0.05	< 0.05
Phenolics	0.08	0.079	0.191	0.075
Phosphorus	0.75	1.2	1.0	1.1
Total Dissolved Solids	4,800	5,700	5,600	4,100
Total Organic Carbon	170	580	560	110
Sulfate	180	240	91	83
Other Indicators				
Specific Conductance (umhos/cm)	n/a	n/a	8.41	7.14
Turbidity (NTU)	n/a	n/a	n/a	n/a
pH (pH units)	n/a	n/a	7.00	6.98
Redox potential (mV)	n/a	n/a	n/a	n/a

Notes:

mg/L = milligrams per liter.

ND = Not detected above MDL.

NS = Not sampled for analyte

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are indicated in bold

The laboratory report lists all parameters analyzed.

N/A=Parameter was not analyzed during this event

Wexford County Landfill

Leachate Sampling

PCBs by GC/ECD (ug/L)	Sample Identification
	Leachate Tank January 2007
Aroclor 1016	< 0.20
Aroclor 1221	< 0.20
Aroclor 1232	< 0.20
Aroclor 1242	< 0.20
Aroclor 1248	< 0.20
Aroclor 1254	< 0.20
Aroclor 1260	< 0.20
Aroclor 1262	< 0.20
Aroclor 1268	< 0.20

Notes ug/L = micrograms per liter.
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NS = Not sampled for analyte
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VOCs = Volatile organic compounds
N/A=Parameter was not analyzed during this event

Wexford County Landfill

Leachate Sampling

Pesticides by GC/ECD (ug/L)	Sample Identification
	Leachate Tank January 2007
4,4'-DDD	< 0.20
4,4'-DDE	< 0.20
4,4'-DDT	< 0.20
Aldrin	< 0.10
alpha-BHC	< 0.10
alpha-Chlordane	< 0.50
beta-BHC	< 0.10
Chlordane (Technical)	< 0.50
delta-BHC	< 0.10
Dieldrin	< 0.20
Endosulfan I	< 0.30
Endosulfan II	< 0.30
Endosulfan sulfate	< 0.50
Endrin	< 0.20
Endrin aldehyde	< 0.20
Endrin ketone	< 0.20
gamma-BHC	< 0.10
gamma-Chlordane	< 0.50
Heptachlor	< 0.10
Heptachlor epoxide	< 0.10
Methoxychlor	< 5.0
Toxaphene	< 10

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 VOCs = Volatile organic compounds
 N/A=Parameter was not analyzed during this event

Wexford County Landfill

Leachate Sampling

Semivolatiles by GC/MS (ug/L)	Sample Identification
	Leachate Tank January 2007
1,2,4-Trichlorobenzene	< 5.0
1,2-Dichlorobenzene	< 5.0
1,2-Diphenylhydrazine	< 5.0
1,3-Dichlorobenzene	< 5.0
1,4-Dichlorobenzene	< 5.0
2,4,5-Trichlorophenol	< 50
2,4,6-Trichlorophenol	< 5.0
2,4-Dichlorophenol	< 5.0
2,4-Dimethylphenol	< 5.0
2,4-Dinitrophenol	< 20
2,4-Dinitrotoluene	< 5.0
2,6-Dichlorophenol	< 5.0
2,6-Dinitrotoluene	< 5.0
2-Chloronaphthalene	< 5.0
2-Chlorophenol	< 5.0
2-Methylnaphthalene	< 5.0
2-Methylphenol	< 5.0
2-Nitroaniline	< 20
2-Nitrophenol	< 5.0
3 & 4-Methylphenol	100
3,3'-Dichlorobenzidine	< 20
3-Nitroaniline	< 20
4,6-Dinitro-2-methylphenol	< 20
4-Bromophenyl phenyl ether	< 5.0
4-Chloro-3-methylphenol	< 5.0
4-Chloroaniline	< 20
4-Chlorophenyl phenyl ether	< 5.0
4-Nitroaniline	< 20
4-Nitrophenol	< 20
Acenaphthene	< 5.0
Acenaphthylene	< 5.0
Aniline	< 20
Anthracene	< 5.0
Benzidine	< 50
Benzo[a]anthracene	< 1.0
Benzo[a]pyrene	< 2.0
Benzo[b]fluoranthene	< 2.0
Benzo[g,h,i]perylene	< 5.0
Benzo[k]fluoranthene	< 5.0
Benzoic acid	120
Benzyl alcohol	< 50

Wexford County Landfill

Leachate Sampling

Semivolatiles by GC/MS (ug/L)	Sample Identification
	Leachate Tank January 2007
Bis(2-chloroethoxy)methane	< 5.0
Bis(2-chloroethyl)ether	< 1.0
Bis(2-chloroisopropyl)ether	< 5.0
Bis(2-ethylhexyl)phthalate	< 5.0
Butyl benzyl phthalate	< 5.0
Chrysene	< 5.0
Dibenz[a,h]anthracene	< 2.0
Dibenzofuran	< 5.0
Diethyl phthalate	< 5.0
Dimethyl phthalate	< 5.0
Di-n-butyl phthalate	< 5.0
Di-n-octyl phthalate	< 5.0
Diphenylamine	< 5.0
Fluoranthene	< 5.0
Fluorene	< 5.0
Hexachlorobenzene	< 5.0
Hexachlorobutadiene	< 5.0
Hexachlorocyclopentadiene	< 5.0
Hexachloroethane	< 5.0
Indeno[1,2,3-cd]pyrene	< 2.0
Isophorone	< 5.0
Naphthalene	< 5.0
Nitrobenzene	< 5.0
N-Nitrosodimethylamine	< 5.0
N-Nitrosodi-n-propylamine	< 5.0
N-Nitrosodiphenylamine	< 5.0
Pentachlorophenol	< 20
Phenanthrene	< 5.0
Phenol	7.4
Pyrene	< 5.0
Pyridine	< 5.0

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Wexford County Landfill Leachate Sampling

Others (mg/L)	Sample Identification
	Leachate Tank January 2007
BOD, 5 Day, 20° C	140
CBOD, 5 Day, 20° C	140
Suspended Solids	52

Notes mg/L = milligrams per liter.
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 Compounds which exceeded Method Detection Limits (MDL)
 are indicated in bold.
 The laboratory report lists all parameters analyzed.